

Einflussfaktoren auf Bodengesundheit

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Projektaufgaben

Bodengesundheit mit ML bewerten

Konzeption & Planung

- Literaturrecherche zu Indikatoren für Bodengesundheit
- Auswahl einer oder mehrerer Bodeneigenschaften (Indikator)
- Identifikation & Auswahl möglicher Einflussfaktoren
- Konkretisierung der Fragestellung
- Datenquellenrecherche & Datenauswahl
- Algorithmenrecherche /-wahl
- Festlegung der Methodik

Modellierung & Analyse

- Untersuchung der Effekte der Einflussfaktoren auf Indikator
- Identifikation der relativen Wichtigkeit der Einflussfaktoren
- Untersuchung von Wechselwirkungen
- Vergleich veränderter Parameter
- räumliche Untersuchung(en) (z.B. Vergleich von Regionen, Skalen, Klimazonen...)

Interpretation & Darstellung

- Welche Faktoren sind am wichtigsten? Welche sind positiv? Welche negativ?
- Gibt es Interaktionen?
- Gibt es lokale / regionale Unterschiede? Wieso?
- Was erwartet man bei künftigen Änderungen (z.B. Klimawandel / Landnutzung)?
- Was fördert Bodengesundheit?
- Visualisierung (Karten, Tabellen, Graphiken...)



Machine learning: A tool for studying soil complexity

Machine Learning and Data Mining

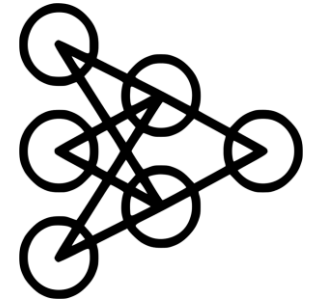
Studying complex relationships requires tools capable of dealing with complexity:

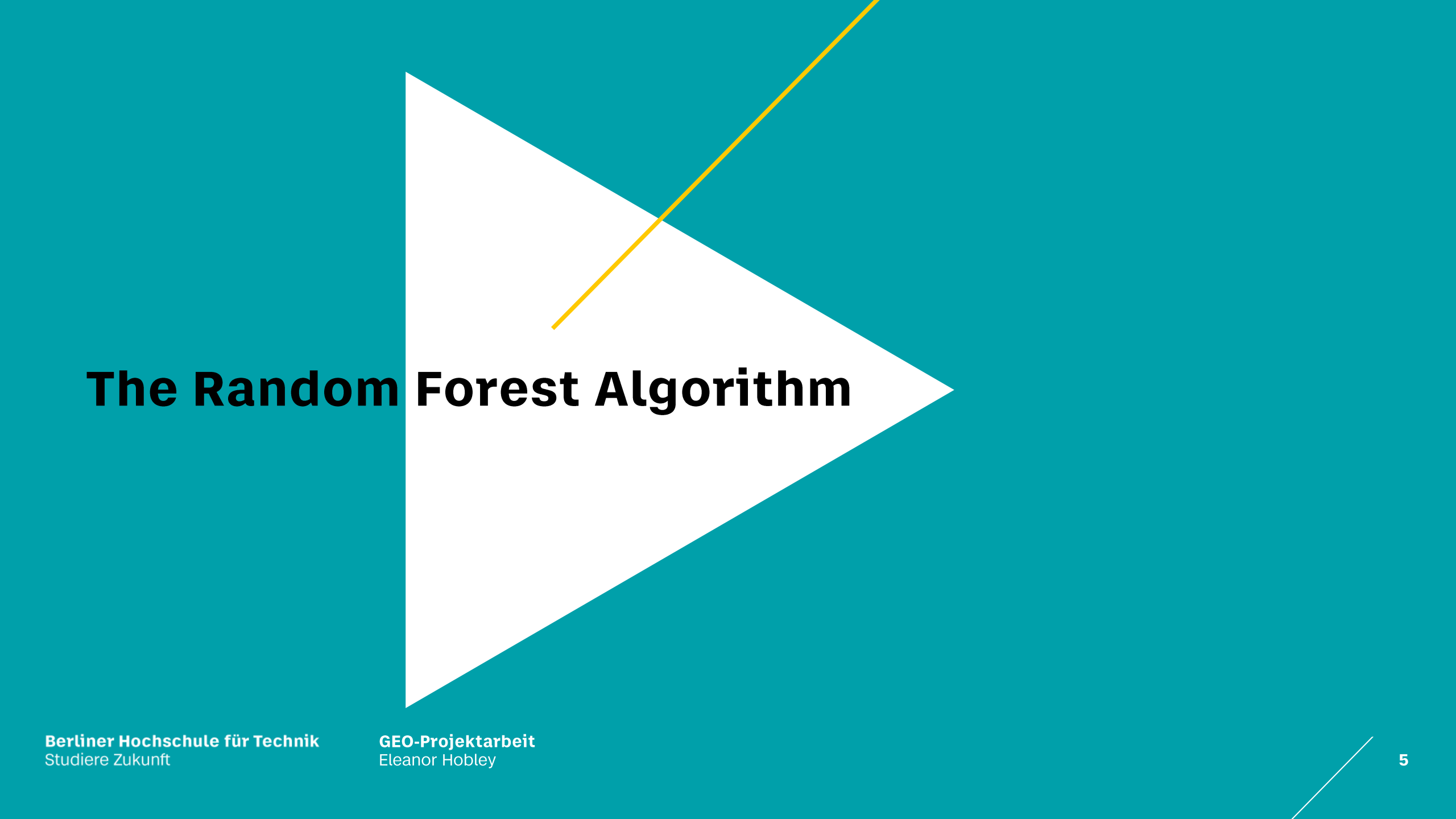
Machine Learning (ML)

Algorithms learn from complex data without prior instruction (assumptions) as to relationships in data

Data mining (DM)

Investigation of relationships in complex datasets via modelling with ML and interpretation of models





The Random Forest Algorithm

What is a Random Forest?

Algorithm to model complex data

Aim: Model (complex) data, $y \sim f(x_1, x_2, x_3 \dots)$ without assumptions of linearity

- Random Forests (2001) Leo Breiman, Machine Learning, [doi: 10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324)
- Used for regression and classification models
- Stochastic (bootstrapped) ensemble of decision trees with aggregation of results across trees
- Bootstrapped aggregation: 'bagging'
- Doesn't overfit 😊

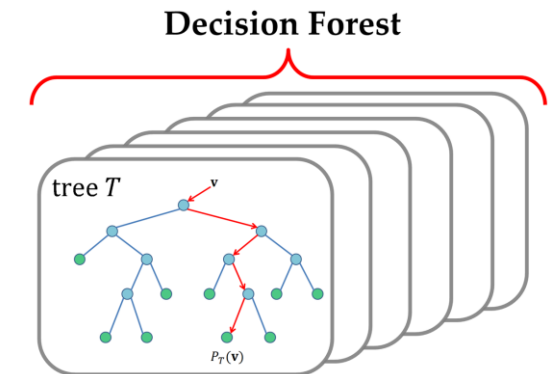


Image: iq.opengenus.org/random-decision-forest/

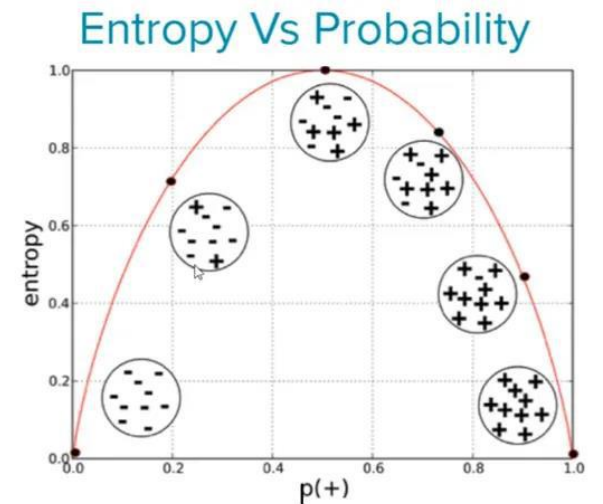
What is a decision tree?

Simple model to partition data in similar groups

- Observations: $y_1 \dots y_n$, dependant variables
- Predictors: $x_1 \dots x_m$, independant variables (features)
- Recursive partitioning structure

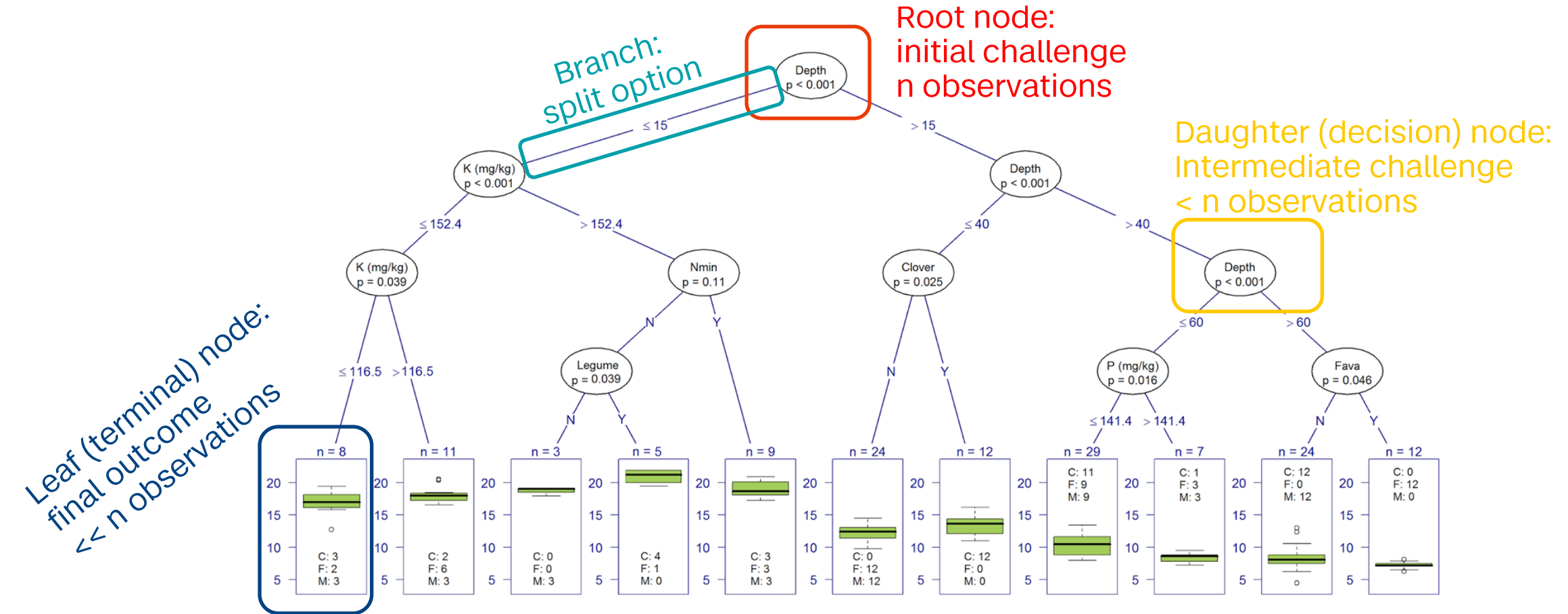
Steps

1. Find predictor such that, for a given value of x , y is partitioned (split) into two subsets (nodes) with maximum group purity (minimum entropy, RMSE)
2. Split y binarily into two daughter nodes at selected value of x
3. Repeat recursively



<https://medium.com/@ashisharora2204/decision-tree-in-pursuit-of-purity-6b1506f0fb15>

What is a decision tree?



Conditional inference tree shows which management options lead to an increase in C-storage in different soil depths

<https://www.sciencedirect.com/science/article/pii/S016788091830255X>

From decision trees to Random Forests

The RF Algorithm

- Select a subset of observations (y_{train}) *at random with replacement* (bootstrap)
- Select a random subset of predictors (x_{train})
- Fit a decision tree with training data:
$$y_{\text{train}} \sim f(x_{\text{train}})$$
- Repeat for 'desired' number of trees
- Predict results of each tree individually for the unselected (out-of-bag) observations (y_{OOB})
- Aggregate (average) results to obtain final predictions per observation

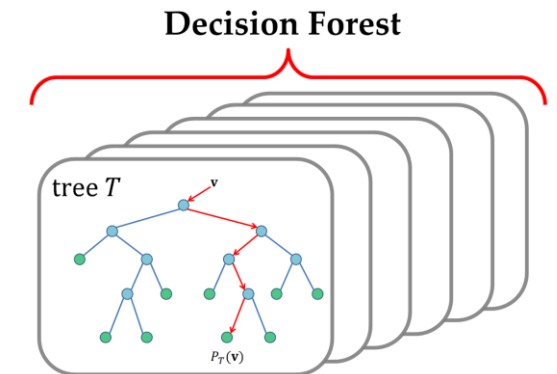


Image: iq.opengenus.org/random-decision-forest/

Example of data selection for a single tree

Random Forest Training Data Selection

Randomly selected predictors

Randomly selected observations

OOB observations

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Depth	POINTID	pH_CaCl2	pH_H2O	EC	OC	CaCO3	P	N	K	NUTS_0	NUTS_1	NUTS_2	NUTS_3	TH_LAT	TH_LONG	SURVEY_DATE	Elev	LC0_Desc
1	0-20 cm	47862690	4.1	4.81	8.73	12.4	3	< LOD	1.1	101.9	AT	AT1	AT11	AT113	47.15024	16.13421	6/07/2018	291	Woodland
2	0-20 cm	47882704	4.1	4.93	5.06	16.7	1	< LOD	1.3	51.2	AT	AT1	AT11	AT113	47.27427	16.17536	6/07/2018	373	Woodland
3	0-20 cm	47982688	4.1	4.85	12.53	47.5	1	12.3	3.1	114.8	AT	AT1	AT11	AT113	47.12326	16.28969	2/06/2018	246	Woodland
4	0-20 cm	48022702	5.5	5.8	21.1	28.1	3	< LOD	2	165.8	AT	AT1	AT11	AT113	47.24569	16.35751	6/07/2018	305	Woodland
5	0-20 cm	48062708	6.1	6.48	10.89	19.4	2	< LOD	2.2	42.1	AT	AT1	AT11	AT113	47.29637	16.41678	5/07/2018	335	Woodland
6	0-20 cm	48122730	6.3	6.81	14.46	18	NA	68	2	277.2	AT	AT1	AT11	AT111	47.48882	16.52059	18/06/2018	232	Cropland
7	0-20 cm	48202738	5.8	6.37	17.08	12.5	NA	58.1	1.4	315.5	AT	AT1	AT11	AT111	47.5541	16.63556	18/06/2018	283	Cropland
8	0-20 cm	48222780	7.1	7.98	23.55	22.9	96	17.5	2.1	329.1	AT	AT1	AT11	AT112	47.9287	16.71116	21/05/2018	136	Cropland
9	0-20 cm	48342774	7.4	8.09	22.61	29.1	86	62.2	2.9	367.8	AT	AT1	AT11	AT112	47.86506	16.86385	26/06/2018	115	Cropland
10	0-20 cm	48442784	7.2	7.84	14.94	14.4	3	< LOD	1.6	146.2	AT	AT1	AT11	AT112	47.94616	17.00926	11/06/2018	166	Cropland
11	0-20 cm	48462776	7.6	8.15	30.9	21.8	112	22.3	2	220.2	AT	AT1	AT11	AT112	47.87283	17.02603	14/05/2018	140	Cropland
12	0-20 cm	48482762	7.4	8.3	23.16	30	271	28.4	2.7	323.3	AT	AT1	AT11	AT112	47.74579	17.03537	3/07/2018	118	Cropland
13	0-20 cm	46562790	7.1	7.52	14.82	15.2	1	44.5	1.9	134.4	AT	AT1	AT12	AT121	48.13165	14.50151	7/06/2018	355	Cropland
14	0-20 cm	46562788	5.2	5.82	14.37	37.3	NA	20.2	4.2	115.6	AT	AT1	AT12	AT121	48.03631	14.62739	18/05/2018	341	Grassland
15	0-20 cm	46662794	3.9	4.74	12.06	41.2	NA	19.4	3.8	294.1	AT	AT1	AT12	AT121	48.162	14.63887	5/06/2018	269	Woodland
16	0-20 cm	46722786	6.4	6.8	15.52	16.9	NA	53.3	2.1	217.6	AT	AT1	AT12	AT121	48.08676	14.71267	5/06/2018	352	Cropland

<https://publications.jrc.ec.europa.eu/repository/handle/JRC129926>

Questions?

Random Forest Training Data Selection

Randomly selected predictors

Randomly selected observations

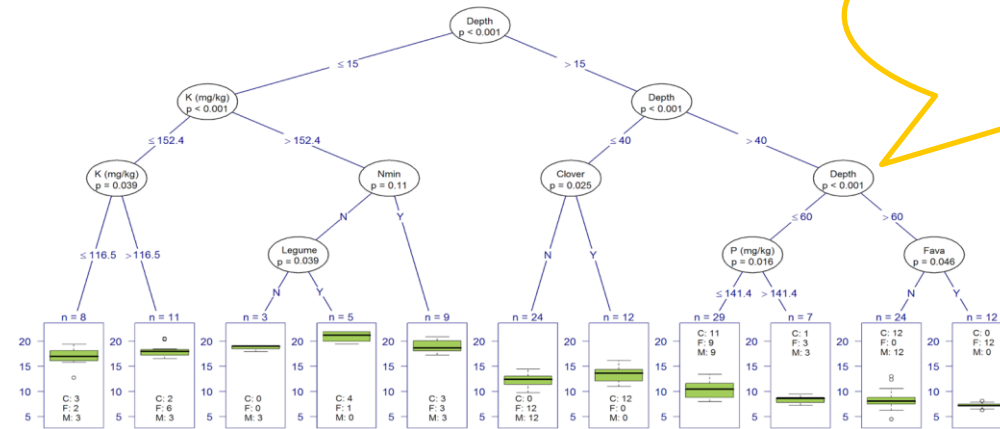
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Questions?

How do we know when to stop splitting?



How many observations should be randomly selected?

How many trees are needed?

Random Forest Training Data Selection

Randomly selected observations

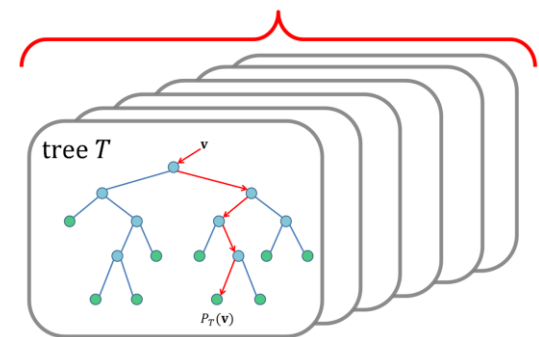
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How many predictors should be randomly selected?

Decision Forest



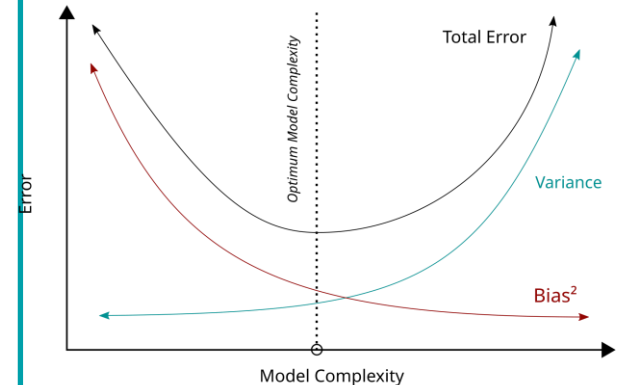
Model optimisation

Challenge

- Models can be very large and slow to fit (and predict)
- Complexity $\sim$ number of trees
- If trees are split indefinitely, each leaf node could result in single observations
- Size of subset y_{train} influences optimal number of trees
- Number of predictors influences optimal number of trees



Optimise model parameters to address these challenges



https://en.wikipedia.org/wiki/Bias%E2%80%93variance_tradeoff#/media/File:Bias_and_variance_contributing_to_total_error.svg

(Some) model parameters for optimisation

(Party package in R)

<https://cran.r-project.org/web/packages/party/party.pdf>

Basic Parameters

Parameter	Meaning
ntree	number of trees in model
mtry	number of predictors selected for training
fraction	proportion of bootstrapped observations per tree

Parameters to stop tree growth

Parameter	Meaning
minsplit	minimum number of observations in node to be split
minbucket	minimum number of observations in node after split
mincriterion	P-value associated with split

Questions?

